



Circle: RACE-01

Single Correct Type Questions:

1. Find the equation of the circle that passes through the points $(1, 0)$, $(-1, 0)$ and $(0, 1)$.
 (A) $x^2 + y^2 = 2$ (B) $x^2 + y^2 = 1$
 (C) $x^2 - 2x + y^2 = 0$ (D) $x^2 + 2x + y^2 = 2$
MCR166
2. The radius of the circle passing through the points $(1, 2)$, $(5, 2)$ & $(5, -2)$ is
 (A) $5\sqrt{2}$ (B) $2\sqrt{5}$
 (C) $3\sqrt{2}$ (D) $2\sqrt{2}$
MCR167
3. $ABCD$ is a square in first quadrant whose side is a , taking AB and AD as axes, then the equation to the circle circumscribing the square is
 (A) $x^2 + y^2 = a(x + y)$ (B) $x^2 + y^2 = a^2(x + y)$
 (C) $x^2 + y^2 = 2a(x + y)$ (D) $x^2 + y^2 = \sqrt{2}a(x + y)$
MCR168
4. The centres of the circles $x^2 + y^2 - 6x - 8y - 7 = 0$ and $x^2 + y^2 - 4x - 10y - 3 = 0$ are the ends of the diameter of the circle
 (A) $x^2 + y^2 - 5x - 9y + 26 = 0$ (B) $x^2 + y^2 + 5x - 9y + 14 = 0$
 (C) $x^2 + y^2 + 5x - y - 14 = 0$ (D) $x^2 + y^2 + 5x + y + 14 = 0$
MCR169
5. The circle described on the line joining the points $(0, 1)$, (a, b) as diameter cuts the x -axis in points whose abscissa are roots of the equation:
 (A) $x^2 + ax + b = 0$ (B) $x^2 - ax + b = 0$
 (C) $x^2 + ax - b = 0$ (D) $x^2 - ax - b = 0$
MCR170
6. A point $A(2, 1)$ is outside the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ & AP, AQ are tangents to the circle. The equation of the circle circumscribing the triangle APQ is :
 (A) $(x + g)(x - 2) + (y + f)(y - 1) = 0$
 (B) $(x + g)(x - 2) - (y + f)(y - 1) = 0$
 (C) $(x - g)(x + 2) + (y - f)(y + 1) = 0$
 (D) $(x - g)(x - 2) + (y - f)(y - 1) = 0$
MCR171
7. Find the equation to the circle which passes through the origin and cuts off intercepts equal to 3 & 4 from the positive axes.
 (A) $x^2 + y^2 + 3x + 4y = 0$ (B) $x^2 + y^2 - 3x + 4y = 0$
 (C) $x^2 + y^2 + 3x - 4y = 0$ (D) $x^2 + y^2 - 3x - 4y = 0$
MCR172

8. Find equation of circle which touches x & y axis & perpendicular distance of centre of circle from $3x + 4y + 11 = 0$ is 5. Given that circle lies in Ist quadrant.
 (A) $x^2 + y^2 + 4x + 4y + 4 = 0$ (B) $x^2 + y^2 - 4x - 4y + 4 = 0$
 (C) $x^2 + y^2 - 4x - 4y + 5 = 0$ (D) $x^2 + y^2 - 4x - 4y + 10 = 0$ MCR173
9. Find the equation to the circle which touches the axis of x at a distance 3 from the origin and intercepts a distance 6 on the axis of y .
 (A) $x^2 + y^2 \pm 3\sqrt{2}y \pm 6x + 9 = 0$ (B) $x^2 + y^2 \pm 3\sqrt{2}y \pm 3x + 9 = 0$
 (C) $x^2 + y^2 \pm 6\sqrt{2}y \pm 6x + 9 = 0$ (D) $x^2 + y^2 \pm 6\sqrt{2}y \pm 6x + 6 = 0$ MCR174
10. Find equation of circle whose cartesian equation are $x = -3 + 2 \sin \theta, y = 4 + 2 \cos \theta$
 (A) $(x + 3)^2 + (y - 4)^2 = 4$ (B) $(x - 3)^2 + (y - 4)^2 = 5$
 (C) $(x + 3)^2 + (y + 4)^2 = 4$ (D) $(x - 3)^2 + (y - 4)^2 = 5$ MCR175
11. The intercepts made by the circle $x^2 + y^2 - 5x - 13y - 14 = 0$ on the x -axis and y -axis are respectively
 (A) 9, 13 (B) 5, 13 (C) 9, 15 (D) None MCR176

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11
Answer	B	D	A	A	B	A	D	B	C	A	C



Circle: RACE-02

Single Correct Type Questions:

1. Find the points of intersection of the line $x - y + 2 = 0$ and the circle $3x^2 + 3y^2 - 29x - 19y + 56 = 0$. Also determine the length of the chord intercepted.
 (A) $(1, 5), (3, 7), 4\sqrt{2}$ (B) $(1, 3), (5, 7), 4\sqrt{2}$
 (C) $(2, 3), (3, 7), 4\sqrt{2}$ (D) $(1, 3), (7, 5), 4\sqrt{2}$
MCR177
2. Show that the line $7y - x = 5$ touches the circle $x^2 + y^2 - 5x + 5y = 0$ and find the equation of the other parallel tangent.
 (A) $x - 7y - 45 = 0$ (B) $x - 5y - 45 = 0$
 (C) $x - 7y - 35 = 0$ (D) $x - 5y - 35 = 0$
MCR178
3. Find the equation of the normal to the circle $x^2 + y^2 = 2x$, which is parallel to the line $x + 2y = 3$.
 (A) $x - 2y - 1 = 0$ (B) $x + 2y - 1 = 0$
 (C) $x + 2y + 1 = 0$ (D) $x - 2y + 1 = 0$
MCR179
4. The equation of normal to the circle $x^2 + y^2 - 4x + 4y - 17 = 0$ which passes through $(1, 1)$ is
 (A) $3x + y - 4 = 0$ (B) $x - y = 0$
 (C) $x + y = 0$ (D) $3x - y - 4 = 0$
MCR180
5. The normal at the point $(3, 4)$ on a circle cuts the circle at the point $(-1, -2)$. Then the equation of the circle is
 (A) $x^2 + y^2 + 2x - 2y - 13 = 0$
 (B) $x^2 + y^2 - 2x - 2y - 11 = 0$
 (C) $x^2 + y^2 - 2x + 2y + 12 = 0$
 (D) $x^2 + y^2 - 2x - 2y + 14 = 0$
MCR181
6. The tangent lines to the circle $x^2 + y^2 - 6x + 4y = 12$ which are parallel to the line $4x + 3y + 5 = 0$ are given by:
 (A) $4x + 3y - 7 = 0, 4x + 3y + 15 = 0$
 (B) $4x + 3y - 31 = 0, 4x + 3y + 19 = 0$
 (C) $4x + 3y - 17 = 0, 4x + 3y + 13 = 0$
 (D) $4x + 3y - 31 = 0, 4x + 3y - 19 = 0$
MCR182
7. The tangent to the circle $x^2 + y^2 = 5$ at the point $(1, -2)$ also touches the circle $x^2 + y^2 - 8x + 6y + 20 = 0$ at
 (A) $(-2, 1)$ (B) $(-3, 0)$ (C) $(-1, -1)$ (D) $(3, -1)$
MCR183

Multiple Correct Type Questions:

8. The equation of circles passing through $(3, -6)$ touching both the axes is
 (A) $x^2 + y^2 - 6x + 6y + 9 = 0$ (B) $x^2 + y^2 + 6x - 6y + 9 = 0$
 (C) $x^2 + y^2 + 30x - 30y + 225 = 0$ (D) $x^2 + y^2 - 30x + 30y + 225 = 0$ **MCR184**
9. Equations of circles which pass through the points $(1, -2)$ and $(3, -4)$ and touch the x -axis is
 (A) $x^2 + y^2 + 6x + 2y + 9 = 0$ (B) $x^2 + y^2 + 10x + 20y + 25 = 0$
 (C) $x^2 + y^2 - 6x + 4y + 9 = 0$ (D) $x^2 + y^2 + 10x + 20y - 25 = 0$ **MCR185**
10. One of the diameter of the circle circumscribing the rectangle $ABCD$ is $x - 3y + 1 = 0$. If two vertices of rectangle are the points $(-2, 5)$ and $(6, 5)$ respectively, then which of the following hold(s) good?
 (A) Area of rectangle $ABCD$ is 64 square units.
 (B) Centre of circle is $(2, 1)$
 (C) The other two vertices of the rectangle are $(-2, -3)$ and $(6, -3)$
 (D) Equation of sides are $x = -2, y = -3, x = 5$ and $y = 6$. **MCR186**

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	A	B	A	B	B	D	A,D	B,C	A,B,C



Circle: RACE-03

Single Correct Type Questions:

1. Find the values of p for which the power of a point $(2, 5)$ is negative with respect to a circle $x^2 + y^2 - 8x - 12y + p = 0$ which neither touches the axes nor cuts them.
 (A) $P \in (16, 47)$ (B) $P \in (16, 36)$
 (C) $P \in (25, 47)$ (D) $P \in (36, 47)$

MCR187

2. A line segment through a point P cuts a given circle in 2 points A & B such that $PA = 16$ & $PB = 9$, find the length of tangent from points to the circle
 (A) 7 (B) 25 (C) 12 (D) 8

MCR188

3. If the length of the tangent from (f, g) to the circle $x^2 + y^2 = 6$ be twice the length of the tangent from (f, g) to the circle $x^2 + y^2 + 3x + 3y = 0$, then $f^2 + g^2 + 4f + 4g$ is equal to
 (A) -1 (B) -2 (C) 0 (D) None of these

MCR189

4. The length of the tangent drawn from any point on the circle $x^2 + y^2 + 2gx + 2fy + p = 0$ to the circle $x^2 + y^2 + 2gx + 2fy + q = 0$ is
 (A) $\sqrt{q - p}$ (B) $\sqrt{p - q}$ (C) $\sqrt{q + p}$ (D) $\sqrt{2q + p}$

MCR190

5. Find the equation of director circle of the circle $(x + 4)^2 + y^2 = 8$
 (A) $(x - 4)^2 + y^2 = 4$ (B) $(x + 4)^2 + y^2 = 16$
 (C) $(x - 4)^2 + y^2 = 16$ (D) $(x + 4)^2 + y^2 = 4$

MCR191

6. The locus of the point of intersection of the tangents to the circle $x^2 + y^2 = a^2$ at points whose parametric angles differ by $\frac{\pi}{3}$ is

(A) $x^2 + y^2 = \frac{4a^2}{3}$ (B) $x^2 + y^2 = \frac{2a^2}{3}$
 (C) $x^2 + y^2 = \frac{a^2}{3}$ (D) $x^2 + y^2 = \frac{a^2}{9}$

MCR192

Multiple Correct Type Questions:

7. The points on the line $x = 3$ from which the tangents drawn to the circle $x^2 + y^2 = 7$ are at right angles is/are
 (A) $(3, \sqrt{5})$ (B) $(3, -\sqrt{5})$
 (C) $(5, \sqrt{3})$ (D) $(5, -\sqrt{3})$

MCR193

8. The centre of a circle passing through the points $(0, 0)$, $(1, 0)$ & touching the circle $x^2 + y^2 = 9$ is :
 (A) $\left(\frac{3}{2}, \frac{1}{2}\right)$ (B) $\left(\frac{1}{2}, \sqrt{2}\right)$ (C) $\left(\frac{1}{2}, \frac{1}{2}\right)$ (D) $\left(\frac{1}{2}, -\sqrt{2}\right)$

MCR194

9. The equation of the circle which touches both the axes and the line $\frac{x}{3} + \frac{y}{4} = 1$ and lies in the first quadrant is $(x - c)^2 + (y - c)^2 = c^2$ where c is
 (A) 1 (B) 2 (C) 4 (D) 6

MCR195

10. The equation(s) of the tangent at the point $(0, 0)$ to the circle where circle makes intercepts of length $2a$ and $2b$ units on the coordinate axes, is (are) -
 (A) $ax + by = 0$ (B) $ax - by = 0$
 (C) $x = y$ (D) $bx + ay = ab$

MCR196

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	C	B	A	B	A	A,B	B,D	A,D	A,B



Circle: RACE-04

Single Correct Type Questions:

1. Find the equation of the chord of the circle $x^2 + y^2 + 6x + 8y + 9 = 0$ whose middle point is $(-2, -3)$.

(A) $x - y + 3 = 0$ (B) $x + y - 5 = 0$ (C) $x + y + 5 = 0$ (D) $x + y - 3 = 0$

MCR197

2. Tangents are drawn to the circle $x^2 + y^2 = 12$ at the points where it is met by the circle $x^2 + y^2 - 5x + 3y - 2 = 0$; find the point of intersection of these tangents.

(A) $\left(6, -\frac{18}{5}\right)$ (B) $\left(-6, -\frac{18}{5}\right)$ (C) $\left(6, \frac{18}{5}\right)$ (D) $\left(-6, \frac{18}{5}\right)$

MCR198

3. The locus of the centers of the circles such that the point $(2, 3)$ is the mid-point of the chord $5x + 2y = 16$ is

(A) $2x - 5y + 11 = 0$ (B) $2x + 5y - 11 = 0$
(C) $2x + 5y + 11 = 0$ (D) $2x - 5y - 11 = 0$

MCR199

4. The chords of contact of the pair of tangents drawn from each point on the line $2x + y = 4$ to the circle $x^2 + y^2 = 1$ pass through the point

(A) $(1, 2)$ (B) $\left(\frac{1}{2}, \frac{1}{4}\right)$ (C) $(2, 4)$ (D) $(4, 4)$

MCR200

5. The locus of the mid-point of a chord of the circle $x^2 + y^2 = 4$ which subtends a right angle at the origin is

(A) $x + y = 2$ (B) $x^2 + y^2 = 1$ (C) $x^2 + y^2 = 2$ (D) $x + y = 1$

MCR201

6. The angle between the two tangents from the origin to the circle $(x - 7)^2 + (y + 1)^2 = 25$ equals

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{6}$

MCR202

7. Circles $C_1: x^2 + y^2 = 9$ & $C_2: x^2 + y^2 - 4x = 0$. Common chord PQ of C_1 & C_2 intersects the direct common tangent at R . If $RP + RQ = \sqrt{\frac{a}{b}}$, where a & b are relatively prime. Then $(a + b)$ equals

(A) 65 (B) 63 (C) 71 (D) 79

MCR203

Multiple Correct Type Questions:

8. For circle $x^2 + y^2 = 16$. Locus of a point P such that area enclosed by tangents drawn from P to circle and chord of contact is 8 is curve C_1 . Then area of curve C_1 is $16k\pi$ where -
 (A) k is even integer (B) k is a prime number
 (C) k is irrational (D) k is odd integer
9. Find the equations of straight lines which pass through the intersection of the lines $x - 2y - 5 = 0, 7x + y = 50$ & divide the circumference of the circle $x^2 + y^2 = 100$ into two arcs whose lengths are in the ratio 2 : 1.
 (A) $3x - 4y - 25 = 0$ (B) $4x + 3y - 25 = 0$
 (C) $4x - 3y - 25 = 0$ (D) $3x + 4y - 25 = 0$

MCR204

MCR205

Matrix Match Type Questions:

- | 10. | Column - I | Column - II |
|-----|---|-------------|
| (A) | Number of values of a for which the common chord of the circles $x^2 + y^2 = 8$ and $(x - a)^2 + y^2 = 8$ subtends a right angle at the origin is | (p) 0 |
| (B) | The number of circles touching all the three lines $3x + 7y = 2, 21x + 49y = 5$ and $9x + 21y = 0$ are | (q) 2 |
| (C) | The length of common chord of circles $x^2 + y^2 - x - 11y + 18 = 0$ and $x^2 + y^2 - 9x - 5y + 14 = 0$ is | (r) 5 |
| (D) | Number of common tangents of the circles $x^2 + y^2 - 2x = 0$ and $x^2 + y^2 + 6x - 6y + 2 = 0$ is | (s) 3 |

MCR206

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	A	A	B	C	C	D	A,B	C,D	A → q; B → p; C → r; D → s



Circle: RACE-05

Single Correct Type Questions:

1. If $y = 2x$ is a chord of the circle $x^2 + y^2 - 10x = 0$, find the equation of a circle with this chord as diameter.

(A) $x^2 + y^2 - x - 2y = 0$

(B) $x^2 + y^2 - 2x - 4y = 0$

(C) $x^2 + y^2 - x + 2y = 0$

(D) $x^2 + y^2 + 2x - y = 0$

MCR207

2. Find the equation of a circle which touches the line $2x - y = 4$ at the point $(1, -2)$ and passes through $(3, 4)$

(A) $(x - 1)^2 + (y + 2)^2 + 10(2x - y - 4) = 0$

(B) $(x - 1)^2 + (y + 2)^2 + 15(2x - y - 4) = 0$

(C) $(x - 1)^2 + (y + 2)^2 + 20(2x - y - 4) = 0$

(D) $(x - 1)^2 + (y + 2)^2 + 25(2x - y - 4) = 0$

MCR208

3. Find the equation of a circle which touches the line $2x - y = 4$ at the point $(1, -2)$ and radius = 5

(A) $(x - 1)^2 + (y + 2)^2 \pm \sqrt{2}(2x - y - 4) = 0$

(B) $(x - 1)^2 + (y + 2)^2 \pm \sqrt{10}(2x - y - 4) = 0$

(C) $(x - 1)^2 + (y + 2)^2 \pm \sqrt{15}(2x - y - 4) = 0$

(D) $(x - 1)^2 + (y + 2)^2 \pm \sqrt{20}(2x - y - 4) = 0$

MCR209

4. Show that the equation $x^2 + y^2 - 2x - 2\lambda y - 8 = 0$ represents for different values of λ a system of circles passing through two fixed points A and B on the x -axis, and also find the equation of that circle of the system the tangent to which at A and B meet on the line $x + 2y + 5 = 0$.

(A) $x^2 + y^2 - 2x - 6y - 8 = 0$

(B) $x^2 + y^2 - x - 3y - 8 = 0$

(C) $x^2 + y^2 + 2x + 6y - 8 = 0$

(D) $x^2 + y^2 + x + 3y - 8 = 0$

MCR210

5. Find the equation of the circle which passes through the point $(1, 1)$ & which touches the circle $x^2 + y^2 + 4x - 6y - 3 = 0$ at the point $(2, 3)$ on it.

(A) $x^2 + y^2 + x - 6y + 3 = 0$

(B) $x^2 + y^2 + x - 6y - 3 = 0$

(C) $x^2 + y^2 + x - 6y + 3 = 0$

(D) $x^2 + y^2 + x - 3y + 3 = 0$

MCR211

6. Equation of the circle which passes through the point $(-1, 2)$ & touches the circle $x^2 + y^2 - 8x + 6y = 0$ at origin, is -

(A) $x^2 + y^2 - 2x - \frac{3}{2}y = 0$

(B) $x^2 + y^2 + x - 2y = 0$

(C) $x^2 + y^2 + 2x + \frac{3}{2}y = 0$

(D) $x^2 + y^2 + 2x - \frac{3}{2}y = 0$

MCR212

7. Find the equations of smallest circle which passes through the points of intersection of the line $x + y = 1$ and the circle $x^2 + y^2 = 9$.
- (A) $x^2 + y^2 - 2x - 2y - 8 = 0$
 (B) $x^2 + y^2 + 2x - 2y - 8 = 0$
 (C) $x^2 + y^2 - x + y - 8 = 0$
 (D) $x^2 + y^2 - x - y - 8 = 0$

MCR213

Multiple Correct Type Questions:

8. If $(a, 0)$ is a point on a diameter segment of the circle $x^2 + y^2 = 4$, then $x^2 - 4x - a^2 = 0$ has
- (A) exactly one real root in $[-1, 0]$
 (B) Exactly one real root in $[2, 5]$
 (C) distinct roots greater than -1
 (D) Distinct roots less than 5

MCR214

Matrix Match Type Questions:

- | 9. | Column - I | Column - II |
|-----|--|-------------|
| (A) | If director circle of two given circles C_1 and C_2 of equal radii touches each other, then ratio of length of internal common tangent of C_1 and C_2 to their radii equals to | (p) 13 |
| (B) | Let two circles having radii r_1 and r_2 are orthogonal to each other. If length of their common chord is k times the square root of harmonic mean between squares of their radii, then k^4 equals to | (q) 7 |
| (C) | The axes are translated so that the new equation of the circle $x^2 + y^2 - 5x + 2y - 5 = 0$ has no first degree terms and the new equation $x^2 + y^2 = \frac{\lambda^2}{4}$, then the value of λ is | (r) 4 |
| (D) | The number of integral points which lie on or inside the circle $x^2 + y^2 = 4$ is | (s) 2 |

MCR215

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9
Answer	B	C	D	A	A	D	D	A,B,C,D	A → s, B → r, C → q, D → p



Circle: RACE-06

Single Correct Type Questions:

1. Find the equations of the transverse common tangents of the circles $x^2 + y^2 - 2x - 6y + 9 = 0$ & $x^2 + y^2 + 6x - 2y + 1 = 0$
 (A) $y = 0, 4x + 3y = 10$ (B) $x = 0, 4x - 3y = 10$
 (C) $x = 0, 3x + 4y = 10$ (D) $y = 0, 3x + 4y = 10$
MCR216
2. Find the equations to the direct common tangents of the circles $x^2 + y^2 - 2x - 6y + 9 = 0$ & $x^2 + y^2 + 6x - 2y + 1 = 0$
 (A) $y = 4$ and $3y = 4x$ (B) $y = 4$ and $4y = 3x$
 (C) $x = 4$ and $3y = 4x$ (D) $x = 4$ and $4y = 3x$
MCR217
3. Number of common tangents of the circles $(x + 2)^2 + (y - 2)^2 = 49$ and $(x - 2)^2 + (y + 1)^2 = 4$ is
 (A) 0 (B) 1
 (C) 2 (D) 3
MCR218
4. If the length of a common internal tangent to two circles is 7, and that of a common external tangent is 11, then the product of the radii of the two circles is
 (A) 18 (B) 20
 (C) 16 (D) 12
MCR219
5. The length of transverse common tangent of the circles $x^2 + y^2 = 1$ and $(x - h)^2 + y^2 = 1$ is $2\sqrt{3}$, then the value of h is :
 (A) ± 2 (B) ± 4
 (C) $\pm\sqrt{3}$ (D) ± 6
MCR220
6. If d denotes the distance of origin from the common tangent to circles S_1 & S_2 where $S_1: x^2 + y^2 - 2x - 4y = 44$ & $S_2: x^2 + y^2 - 8x - 12y + 48 = 0$, then $[d]$ is equal to (where $[\cdot]$ denotes greatest integer function)
 (A) 3 (B) 6
 (C) 9 (D) 10
MCR221
7. Circle K is circle of largest radius, inscribed in the first quadrant touching the circle $x^2 + y^2 = 36$ internally. The length of the radius of the circle K , is -
 (A) $\frac{6 - \sqrt{2}}{2}$ (B) $\frac{3\sqrt{2}}{2}$
 (C) 3 (D) $6(\sqrt{2} - 1)$
MCR222

Multiple Correct Type Questions:

8. For the circles $x^2 + y^2 - 10x + 16y + 89 - r^2 = 0$ and $x^2 + y^2 + 6x - 14y + 42 = 0$ which of the following is/are true.
- (A) Number of integral values of r are 14 for which circles are intersecting.
 (B) Number of integral values of r are 9 for which circles are intersecting.
 (C) For r equal to 13 number of common tangents are 3.
 (D) For r equal to 21 number of common tangents are 2.

MCR223

9. Which of the following statement(s) is/are correct with respect to the circles $S_1 \equiv x^2 + y^2 - 4 = 0$ and $S_2 \equiv x^2 + y^2 - 2x - 4y + 4 = 0$?
- (A) S_1 and S_2 intersect at an angle of 90° .
 (B) The point of intersection of the two circle are $(2, 0)$ and $\left(\frac{6}{5}, \frac{8}{5}\right)$.
 (C) Length of the common of chord of S_1 and S_2 is $\frac{4}{\sqrt{5}}$.
 (D) The point $(2, 3)$ lies outside the circles S_1 and S_2 .

MCR224**Subjective Type Questions:**

10. A circle $S = 0$ is drawn with its centre at $(-1, 1)$ so as to touch the circle $x^2 + y^2 - 4x + 6y - 3 = 0$ externally. Find the intercept made by the circle $S = 0$ on the coordinate axes.

MCR225**ANSWER KEY**

Question	1	2	3	4	5	6	7	8	9	
Answer	C	A	B	A	B	C	D	A,C	A,C,D	



Circle: RACE-07

Single Correct Type Questions:

1. The circles $x^2 + y^2 - 2x - 6y - 12 = 0$ and $x^2 + y^2 + 6x + 4y - 6 = 0$ cut each other at
(A) 60° (B) 30° (C) 45° (D) 90° MCR226
2. Find the equation of the circle passing through the origin and cutting the circles $x^2 + y^2 - 4x + 6y + 10 = 0$ and $x^2 + y^2 + 12y + 6 = 0$ at right angles.
(A) $(x^2 + y^2) - 7x + 2y = 0$ (B) $2(x^2 + y^2) - 7x + 2y = 0$
(C) $(x^2 + y^2) - 14x + 4y = 0$ (D) $2(x^2 + y^2) + 7x - 2y = 0$ MCR227
3. Given the three circles $x^2 + y^2 - 16x + 60 = 0$, $3x^2 + 3y^2 - 36x + 81 = 0$ and $x^2 + y^2 - 16x - 12y + 84 = 0$, find the point from which the tangents to them are equal in length and find this length.
(A) $\left(\frac{33}{4}, 2\right); L_T = \frac{1}{4}$ (B) $\left(\frac{33}{2}, 2\right); L_T = \frac{1}{2}$
(C) $\left(\frac{33}{4}, 4\right); L_T = \frac{1}{4}$ (D) $\left(\frac{11}{4}, 2\right); L_T = \frac{1}{2}$ MCR228
4. Equation of the circle cutting orthogonally the three circles $x^2 + y^2 - 2x + 3y - 7 = 0$, $x^2 + y^2 + 5x - 5y + 9 = 0$ and $x^2 + y^2 + 7x - 9y + 29 = 0$ is
(A) $x^2 + y^2 - 16x - 18y - 4 = 0$ (B) $x^2 + y^2 - 7x + 11y + 6 = 0$
(C) $x^2 + y^2 + 2x - 8y + 9 = 0$ (D) $x^2 + y^2 + 16x - 18y - 4 = 0$ MCR229
5. The line $x + 2y + 3 = 0$ cuts the circle $x^2 + y^2 + 4x + 4y - 1 = 0$ at points P and Q and the line $2x + 3y + \lambda = 0$ cuts the circle $x^2 + y^2 + 6x + 2y - 7 = 0$ at points R and S . If P, Q, R, S are concyclic, then the value of λ is
(A) 2 (B) 4 (C) 6 (D) 8 MCR230

Comprehension Type Questions:

Comprehension # 1 (Q. No. 6 to 8)

Let S_1, S_2, S_3 be the circles $x^2 + y^2 + 3x + 2y + 1 = 0$, $x^2 + y^2 - x + 6y + 5 = 0$ and $x^2 + y^2 + 5x - 8y + 15 = 0$, then

6. Point from which length of tangents to these three circles is same is
(A) (1, 0) (B) (3, 2)
(C) (10, 5) (D) (-2, 1) MCR231

7. Equation of circle S_4 which cut orthogonally to all given circle is
 (A) $x^2 + y^2 - 6x + 4y - 14 = 0$ (B) $x^2 + y^2 + 6x + 4y - 14 = 0$
 (C) $x^2 + y^2 - 6x - 4y + 14 = 0$ (D) $x^2 + y^2 - 6x - 4y - 14 = 0$

MCR232

8. Radical centre of circles S_1, S_2 , & S_4 is

- (A) $\left(-\frac{3}{5}, -\frac{8}{5}\right)$ (B) (3, 2) (C) (1, 0) (D) $\left(-\frac{4}{5}, -\frac{3}{2}\right)$

MCR233

Comprehension # 2 (Q. No. 9 to 11)

Two circles are

$$S_1 \equiv (x + 3)^2 + y^2 = 9$$

$$S_2 \equiv (x - 5)^2 + y^2 = 16$$

with centres C_1 & C_2

9. A direct common tangent is drawn from a point P (on x -axis) which touches S_1 & S_2 at Q & R , respectively. Find the ratio of area of ΔPQC_1 & ΔPRC_2 .

- (A) 3 : 4 (B) 9 : 16 (C) 16 : 9 (D) 4 : 3

MCR234

10. From point ' A ' on S_2 which is nearest to C_1 , a variable chord is drawn to S_1 . The locus of mid point of the chord.

- (A) circle (B) Diameter of s_1
 (C) Arc of a circle (D) chord of s_1 but not diameter

MCR235

11. Locus obtained in question above cuts the circle S_1 at B & C , then line segment BC subtends an angle on the major arc of circle S_1 is

- (A) $\cot^{-1} \frac{3}{4}$ (B) $\sin^{-1} \frac{3}{4}$
 (C) $\tan^{-1} \frac{3}{4}$ (D) $\cot^{-1} \frac{3}{4}$

MCR236

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11
Answer	D	B	A	A	B	B	D	A	B	C	A